



CSE370 : Database Systems
Project Report
Project Title : University Portal

Group No : 01, CSE370 Lab Section : 15, Spring 2026		
ID	Name	Contribution
24101352	Mohammad Faiyaz Mazumder	Features Planning,ER, Schema Diagram, Normalization, Github, Database (SQL QUERIES AND TABLE) and Portal Features Coding, Security, Report
23201493	Fabiha Tarannum	ER, Schema Table, SQL queries , Github applications,Features Coding
23201558	Shuvo Das	ER Schema Diagram Normalization, Features Coding, Security

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Introduction

[Briefly describe the idea of your project]

The “**BracU**” **University Portal** is a web-based academic management system designed to digitize and streamline the day-to-day activities of university students. The portal serves as a single platform where students can manage their entire academic life from enrolling in courses to tracking grades, filing complaints, sharing notes, and earning rewards.

Project Features

ID, Name	Features [3 per member]	
24101352, Mohammad Faiyaz Mazumder	Ft 1	Smart Course Recommendation System (Connected to course Enrollment)
	Ft 2	Campus Facility Complaint System
	Ft 3	Reward Points System
23201493, Fabiha Tarannum	Ft 1	Dashboard.
	Ft 2	Student performance analytic dashboard.
	Ft 3	Course enrollment system.
23201558 Shuvo Das	Ft 1	Lab Availability Checker
	Ft 2	Messaging
	Ft 3	Notes Sharing

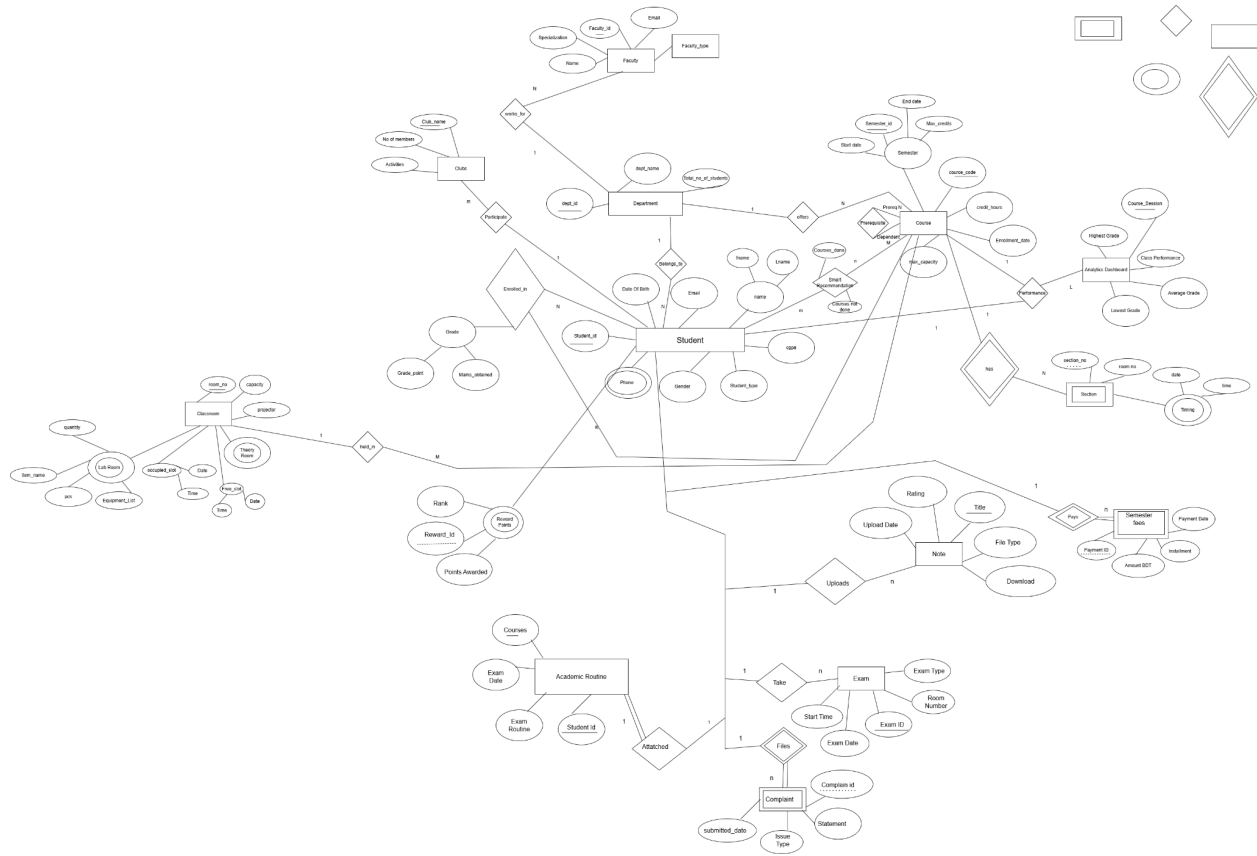
Extra features:

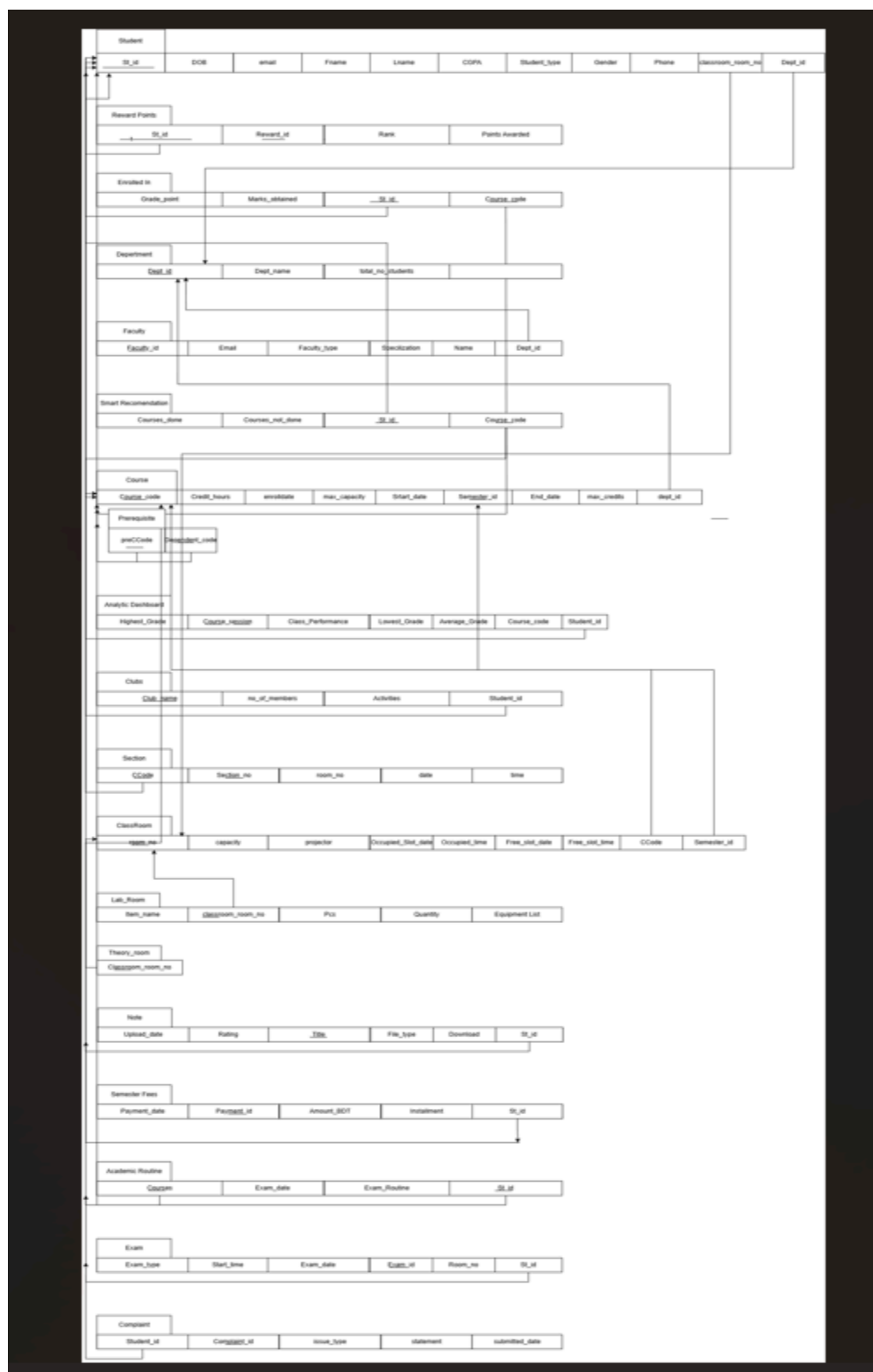
Login and Register (Faiyaz)

Department Stats(Fabiha)

GPA Predictor (Shuvo)

ER/EER Diagram





Schema Diagram

Normalization

- a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.

1NF

By removing composite and multivalued such as reward points and section timing and composite like lab room_no and theory_room_no

- b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?

2NF

already all attributes are fully dependent on the primary keys. No partial dependencies

- c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

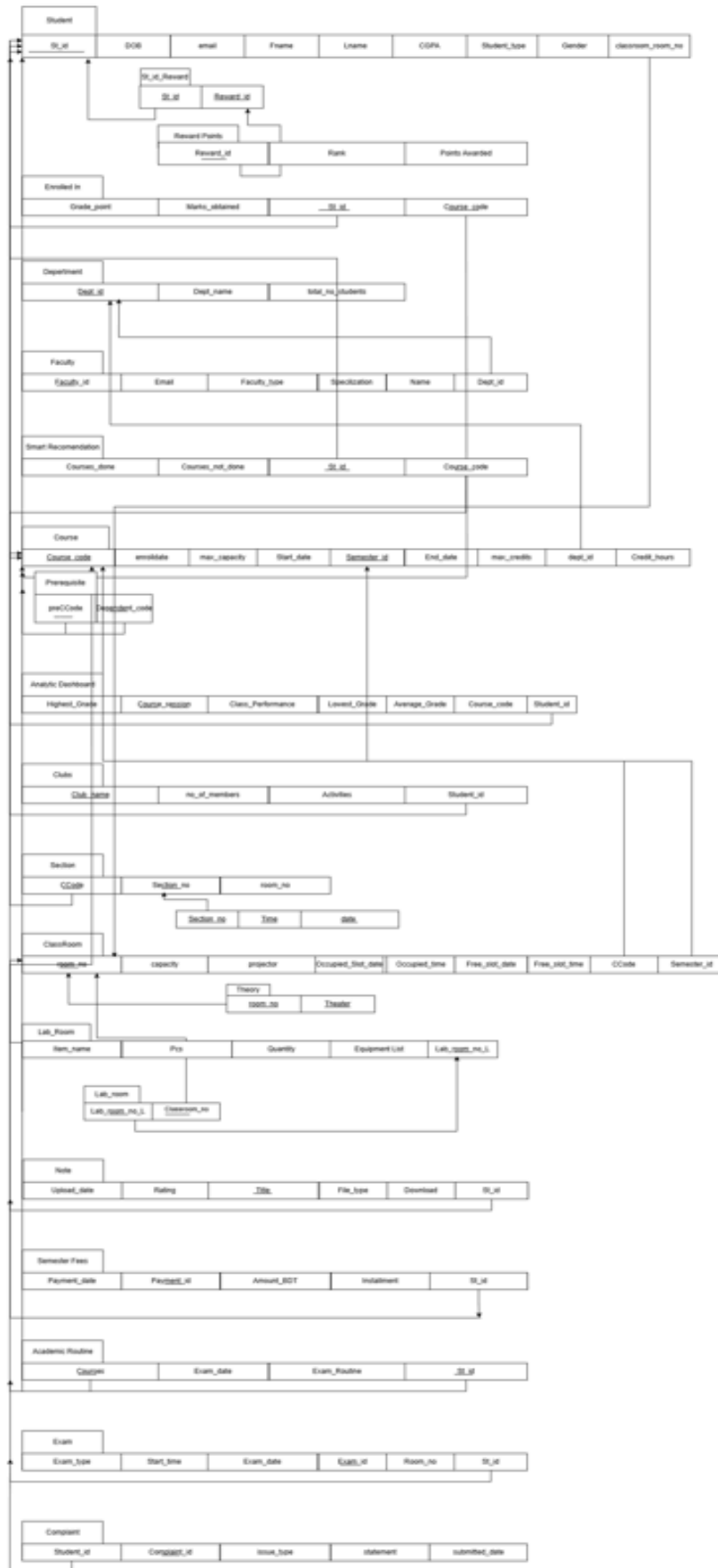
3NF

no transitive dependencies so it is in 3NF

1NF
By removing composite and multivalued such as reward points and section timing and composite the lab room_no and theory_room_no

2NF
Primary attributes are fully dependent on the primary keys. No partial dependencies

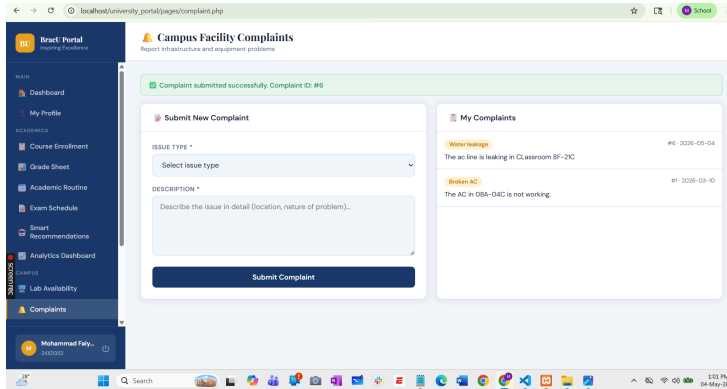
3NF
No transitive dependencies as it is in 2NF



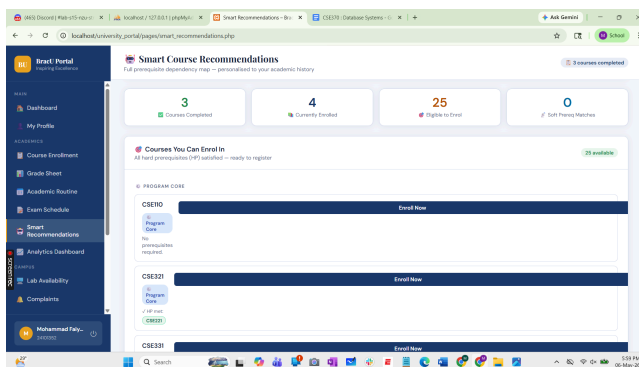
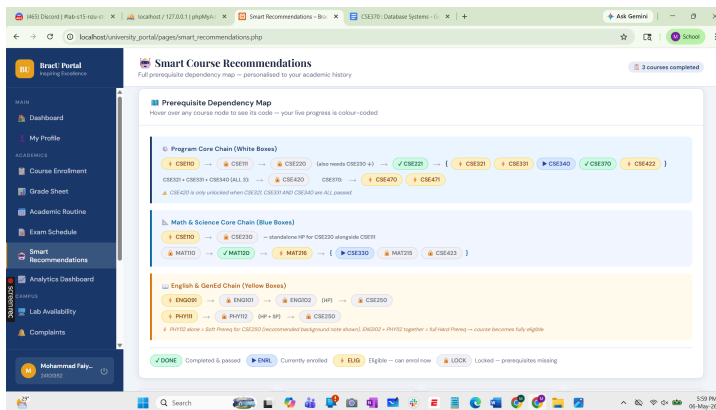
Frontend Development

Contribution of ID : 24101352, Name : Mohammad Faiyaz Mazumder

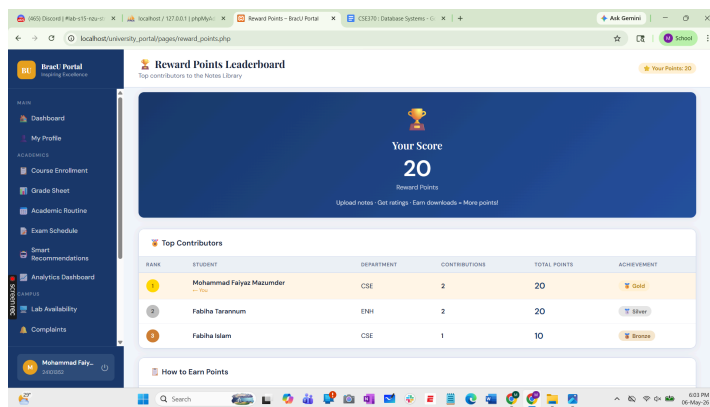
Complaint System



Smart Course Recommendation



Reward Points System



Login

The screenshot shows the login page of the 'Bract Central Portal'. It features a central white box with the portal's logo and name. Below the name is a 'Sign in to your account' section with input fields for 'STUDENT ID OR EMAIL' and 'PASSWORD'. There is a 'Forgot Password?' link and a 'Sign In' button. Below the sign-in section is a 'Register Here' link. At the bottom, there is a note about using a Student ID as both username and password.

Register

The screenshot shows the 'Student Registration' page of the 'Bract Central Portal'. It features a central white box with the portal's logo and name. Below the name is a 'Create your account' section with a 'Student Registration' title. The registration form includes input fields for 'FIRST NAME', 'LAST NAME', 'STUDENT ID', 'UNIVERSITY EMAIL', 'DATE OF BIRTH', 'GENDER', 'DEPARTMENT', and 'PHONE NUMBER'. There is a 'Create Account' button at the bottom.

Contribution of ID : 23201493, Name : Fabiha Tarannum

The screenshot shows a web browser displaying the BracU Portal dashboard. The browser's address bar shows the URL `localhost/university_portal/pages/dashboard.php`. The dashboard is titled "Welcome back, Faiyaz!" and identifies the user as "Computer Science & Engineering - 23201001 - Spring 2026".

Left Sidebar:

- BracU Portal Inspiring Excellence
- MAIN
 - Dashboard
 - My Profile
- ACADEMICS
 - Course Enrollment
 - Grade Sheet
 - Academic Routine
 - Exam Schedule
 - Smart Recommendations
 - Analytics Dashboard
- CAMPUS
 - Lab Availability
 - Complaints
 - Department Stats
- COMMUNITY
 - Notes Library
- User Profile: Faiyaz Ahmed, 23201001

Main Content Area:

- Top Summary Cards:**
 - 3 Enrolled Courses
 - 3.50 Current CGPA
 - 20 Reward Points
 - Paid Fee Status
- Upcoming Exams:** Your next scheduled exams. View All.

TYPE	DATE	TIME	ROOM
Final	15 Jun 2026	09:00 AM	08A-04C
- Recent Grades:** Your latest academic results. View All.

COURSE	GRADE	MARKS
MAT120	B+	80.00/100
CSE370	A	92.00/100
CSE221	A-	86.00/100
- Quick Actions:**
 - Enroll in Course
 - Get Recommendations
 - Browse Notes
 - File Complaint
 - Check Lab Slots
 - View Fees

Student Dashboard

BU

BracU Portal

Inspiring Excellence

MAIN

Dashboard

My Profile

ACADEMICS

Course Enrollment

Grade Sheet

Academic Routine

Exam Schedule

Smart Recommendations

Analytics Dashboard

CAMPUS

Lab Availability

Complaints

Department Stats

COMMUNITY

Notes Library

Falyaz Ahmed

22301001

Course Enrollment

Add or drop courses for Spring 2026

9/15 Credits

Credit Hours Used

9 / 15

My Enrolled Courses

Currently enrolled this semester

COURSE CODE	CREDITS	DEPT	ROOM	ENROLLED	ACTION
CSE221	3 cr	CSE	07A-03C	2/40	Drop
CSE370	3 cr	CSE	08A-03C	5/40	Drop
MAT120	3 cr	MNS	08A-02C	2/40	Drop

Available Courses

Spring 2026 course catalog

COURSE CODE	CREDITS	DEPARTMENT	ENROLLED/CAPACITY	STATUS	ACTION
ACT201	3 cr	Business & Economics	0/40	Open	Enroll
BIO101	3 cr	Mathematics & Natural Sciences	0/40	Open	
BUS102	3 cr	Business & Economics	2/40	Open	

Course Enrolment

BU

BracU Portal

Inspiring Excellence

MAIN

Dashboard

My Profile

ACADEMICS

Course Enrollment

Grade Sheet

Academic Routine

Exam Schedule

Smart Recommendations

Analytics Dashboard

CAMPUS

Lab Availability

Complaints

Department Stats

COMMUNITY

Notes Library

Falyaz Ahmed

22301001

Analytics Dashboard

Course performance & class-wide statistics

Course Performance Analytics

Faculty view — class-wide grade statistics

COURSE / SECTION	HIGHEST GRADE	LOWEST GRADE	CLASS PERFORMANCE
BUS102-SP26	A	B+	Excellent — High engagement
CSE221-SP26	A	B+	Good — Strong algorithmic performance
CSE370-SP26	A	C	Good — Most students above B+
CSE420-SP26	B+	B	Good — Active participation
MAT120-SP26	A-	C+	Average — Some students struggling

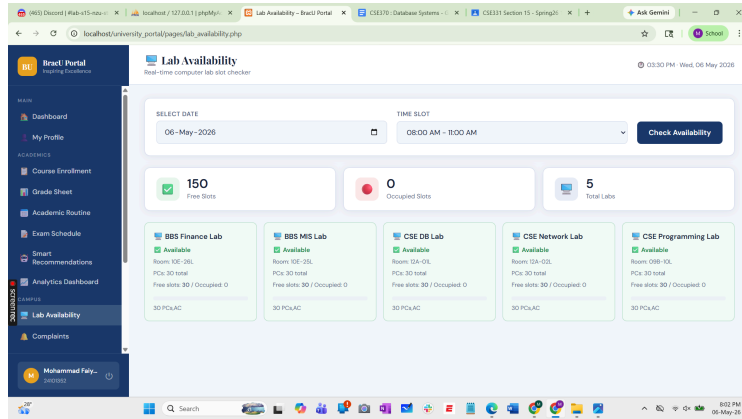
Department Statistics

Real-time enrollment counts by department

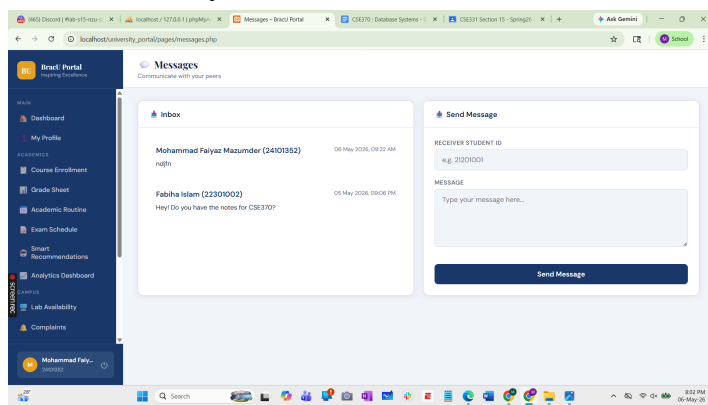
DEPARTMENT	CODE	TOTAL STUDENTS	AVG CGPA	DISTRIBUTION
Computer Science & Engineering	CSE	10	3.49	
Business & Economics	BBS	2	3.65	
Electrical & Electronic Engineering	EEE	2	3.43	
Mathematics & Natural Sciences	MNS	1	3.85	
General Education	GenEd	0	—	

Analytics Dashboard

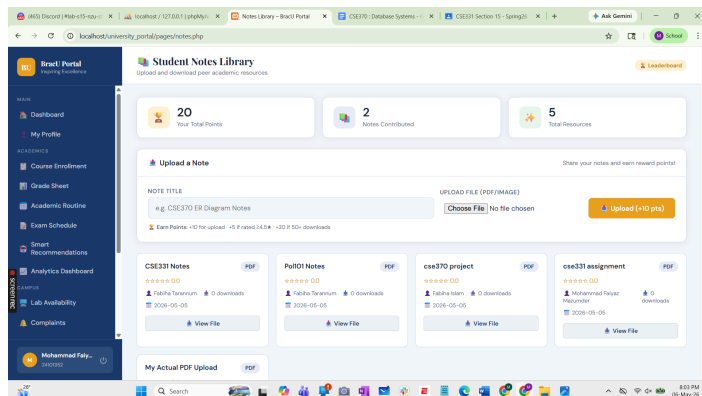
Contribution of ID :23201558, Name : Shuvo Das



Lab Availability Checker



Messaging



Notes Sharing

BI

Bract Portal

Bracting Excellence

FEEDS

Dashboard

My Profile

ACADEMICS

Courses Enrollment

Grade Sheet

Academic Routine

Exam Schedule

Smart Recommendations

Analytics Dashboard

ADVISORS

Lab Availability

Complaints

MO

Muhammad Falye

3800953

Grade Sheet

Academic transcript: Muhammad Falye Meumder

Print

Spring 2026 (Current)

COURSE CODE	CREDIT HOURS	MARKS OBTAINED	GRADE	GRADE POINT	QUALITY POINTS
BI0101	3	--	Pending	--	--
BUS902	3	--	Pending	--	--
CS6330	3	--	Pending	--	--
CS6340	3	--	Pending	--	--
SEMESTER TOTAL	12	--	--	--	--

GPA Predictor

Estimate your future CGPA

Target CGPA: 3.53

COURSE CODE	CREDIT HOURS	PREDICTED GRADE
BI0101	3	A- (3.7)
BUS902	3	A (4.0)
CS6330	3	B- (2.7)
CS6340	3	B+ (3.3)

GPA Calculator

Backend Development

Briefly discuss about Backend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

Contribution of ID : 24101352, Name : Mohammad Faiyaz Mazumder

1) Complaint

Fetching Student's Own Complaints

php

```
$mine = $conn->prepare(
    "SELECT * FROM Complaint WHERE Student_id=24101352 ORDER BY
    submitted_date DESC"
);
```

- Only fetches complaints belonging to the current student
- Ordered by newest first

2) Smart Recommendation System

Build the Prerequisite Map

php

```
$prereq_q = $conn->query("SELECT course_code, prereq_course_code FROM
Prerequisite");
```

```
$prereqs = [];
```

```
while($r = $prereq_q->fetch_assoc()) {
```

```
    $prereqs[$r['course_code']] = $r['prereq_course_code'];
```

```
}
```

- Fetches every prerequisite relationship from the **Prerequisite** table
- Builds a PHP array where each course maps to its list of required prerequisites

Example of what **\$prereqs** looks like in memory:

```
$prereqs = [
```

'CSE370' => ['CSE221'],

'CSE420' => ['CSE221'],

]

3) Reward Points

Used Joins to build the reward system :

\$leaderboard = \$conn->query("

SELECT

s.Student_id, s.Fname, s.Lname, s.dept_id,

SUM(r.Points_Awarded) as total_points,

**(SELECT COUNT(*) FROM Note n WHERE n.Student_id=s.Student_id) as
contributions**

FROM Reward_Points r

JOIN Student s ON r.Student_id=s.Student_id

GROUP BY s.Student_id

ORDER BY total_points DESC

LIMIT 20

");

Backend connected to a faster way to get just your score for the top bar-

**\$my_pts = \$conn->prepare("SELECT SUM(Points_Awarded) as total FROM
Reward_Points WHERE Student_id=?"); \$my_pts->bind_param('s', \$sid);
\$my_pts->execute(); \$my_total = \$my_pts->get_result()->fetch_assoc()['total']**

4) Login and Register

Password Verification

php

```
if ($row = $res->fetch_assoc()) {  
    if ($pw === $sid || password_verify($pw, $pw)) {  
        // login success  
    } else {  
        $error = 'Invalid password.';  
    }  
} else {  
    $error = 'Student ID not found.';  
}
```

- `$res->fetch_assoc()` — fetches the student row if found
- `$pw === $sid` — demo mode: password equals the Student ID (e.g. 22301001)
- `password_verify()` — ready for production use with hashed passwords
- Two separate error messages — one for wrong password, one for Student ID not found

Database Duplicate Check

php

```
$chk = $conn->prepare(  
    "SELECT Student_id FROM Student WHERE Student_id=? OR Email=?"  
);  
  
$chk->bind_param('ss', $sid, $email);  
  
$chk->execute();  
  
if ($chk->get_result()->num_rows > 0) {  
    $error = 'Student ID or Email already exists.';
```

}

- Uses a prepared statement to check both Student ID and Email simultaneously
- **OR** condition catches two cases — someone re-registering same ID or using same email with different ID
- Prevents duplicate records in the database

Contribution of ID : 23201493 Name : Fabiha Tarannum

Authentication & Access Control

```
require_once '../includes/auth.php';  
requireLogin();
```

1. Includes authentication logic from auth.php
2. requireLogin() ensures:

Only logged-in users can access this page

If not logged in → user is redirected (usually to login page)

Database Connection

```
require_once '../includes/db.php';
```

1. Connects PHP to MySQL database using \$conn
2. This connection is used for all queries below

Fetching Course Analytics Data

```
$analytics = $conn->query("SELECT * FROM AnalyticsDashboard ORDER BY  
Course_Section");
```

1. Retrieves all records from AnalyticsDashboard
2. Orders them by Course_Section

Processing Data in PHP

```
$max_s = 1;  
$rows = [];  
while($r = $dept_stats->fetch_assoc()) {
```

```

    $rows[] = $r;
    if($r['total'] > $max_s) $max_s = $r['total'];
}

```

Dynamic Content Generation (PHP inside HTML)

```
<?php while($r = $analytics->fetch_assoc()): ?>
```

What it does:

Dynamically generates table rows from database button

Authentication + Student Identification

```

require_once '../includes/auth.php';
requireLogin();

```

```
$sid = getStudentId();
```

1. **requireLogin()** → ensures only logged-in users can access
2. **getStudentId()** → fetches the currently logged-in student's ID (usually from session)
3. **Counting Enrolled Courses**

```

SELECT COUNT(*) as cnt
FROM Enrolled_In
WHERE Student_id=?

```

1. **Counts how many courses the student is enrolled in**

ENROLL FUNCTION (Main Backend Logic)

```
if ($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST['enroll']))
```

1. **Detects when user clicks Enroll button**

Fetch Student's Enrolled Courses:

```

SELECT e.course_code,c.credit_hours, c.Semester_id,
c.room_no, c.dept_id,
(SELECT COUNT(*) FROM Enrolled_In WHERE course_code=e.course_code) as
enrolled_count,
c.max_capacity
FROM Enrolled_In e
JOIN Course c ON e.course_code=c.course_code
WHERE e.Student_id=?

```

1. **JOIN** → get course details
2. **Subquery** → count students per course

Insert Enrollment

INSERT INTO Enrolled_In (Student_id, course_code)

VALUES (?,?)

Happens only if:

Not already enrolled

Credit limit OK

Capacity available

Contribution of ID: 23201558, Name: Shuvo Das

1. Lab Availability Checker

Real-Time Conflict Detection *[Screenshot: Lab Availability Grid showing Green "Available" and Red "Occupied" cards]*

PHP

\$query = "

SELECT

l.lab_room,

c.pcs,

COUNT(s.section_no) as class_count

FROM Classroom_Lab_Room l

JOIN Classroom c ON TRIM(l.room_no) = TRIM(c.room_no)

LEFT JOIN Section s ON TRIM(l.room_no) = TRIM(s.room_no)

AND s.date = ?

AND s.time >= ?

AND s.time < ?

GROUP BY l.lab_room, c.pcs

";

- Uses a **LEFT JOIN** to ensure all labs are returned, even if no classes are scheduled.
- Implements a specific time-window filter (**>=** start time AND **<** end time) to catch any overlapping 3-hour class blocks.
- Uses **COUNT()** to mathematically determine if the room is occupied (count > 0) or completely free (count = 0).

2. Notes Library & Reward System

Secure File Upload & Automated Gamification *[Screenshot: Upload Note form and "Earn Points" banner]*

PHP

```
// Sanitize filename and prepend timestamp
$safe_title = preg_replace('/[^A-Za-z0-9\-\_]/', '_', $title);
$new_file_name = time() . '_' . $safe_title . '.' . $ext;
$destination = $upload_dir . $new_file_name;

if (move_uploaded_file($file_tmp, $destination)) {
    $ins = $conn->prepare("INSERT IGNORE INTO Note (...) VALUES (?,0,?,0,?,?,?)");
    // Automated Reward System
    if ($ins->execute() && $conn->affected_rows > 0) {
        $rp = $conn->prepare("INSERT INTO Reward_Points (Rank, Points_Awarded,
Student_id) VALUES (0, 10, ?)");
        $rp->bind_param('s', $sid);
        $rp->execute();
    }
}
```

- Uses **preg_replace** to clean special characters and **time()** to guarantee 100% unique filenames, preventing server file overwriting.
- Safely moves the binary file to the **uploads/** directory.
- Automatically triggers a secondary query upon successful upload to instantly award the student 10 Reward Points.

3. Peer-to-Peer Messages

Inbox Management and Auto-Read State *[Screenshot: Messages Inbox showing recent conversations and clear notification badges]*

PHP

```
// Mark messages as read when viewing the inbox
$upd = $conn->prepare("UPDATE Messages SET is_read=TRUE WHERE
receiver_id=?");
$upd->bind_param('s', $sid);
$upd->execute();
```

```
// Fetch inbox messages with Sender details
$inbox_q = $conn->prepare("
    SELECT m.*, s.Fname, s.Lname
    FROM Messages m
    JOIN Student s ON m.sender_id = s.Student_id
    WHERE m.receiver_id = ?
    ORDER BY m.sent_at DESC
");
```

- Executes an immediate **UPDATE** query on page load to clear unread notification badges dynamically.
- Uses a relational **JOIN** to stitch the **Messages** table together with the **Student** table, allowing the UI to display the sender's actual First and Last name rather than just a raw Student ID.

4. GPA Calculator / Predictor

Historical Academic Aggregation [*Screenshot: Academic Transcript table and "Semester GPA" stat card*]

PHP

```
$best_grades = [];
$total_credits = 0;
$total_points = 0;

while($r = $grades_res->fetch_assoc()) {
    $cc = $r['Course_code'];
    if (!isset($best_grades[$cc]) || $r['Grade_Points'] > $best_grades[$cc]['Grade_Points']) {
        $best_grades[$cc] = $r; // Handles retakes by keeping highest grade
    }
}

foreach($best_grades as $g) {
    $total_credits += $g['credit_hours'];
    $total_points += $g['Grade_Points'] * $g['credit_hours'];
}

$gpa = $total_credits > 0 ? round($total_points/$total_credits, 2) : 0;
```

- Loops through historical database records to filter out lower grades in the event of course retakes.
- Calculates total quality points (**Grade_Points * credit_hours**) against total earned credits to establish the current base GPA.
- These backend values are directly injected into the frontend JavaScript engine, allowing the predictor tool to function seamlessly in real-time.

Source Code Repository

Upload source code to GitHub or Google Drive and share a publicly accessible link in this section of the report

Drive

https://drive.google.com/drive/u/0/folders/1a7ileOF_YzTHW1WzavqYPc7tiel_CZ4g

Github

https://github.com/Fabiha-09/CSE370-Database_System-Lab_Project-Group-1--University_Portal

Conclusion

The **BracU University Portal** was developed as a database-driven web application that successfully runs the relational database concepts learned in CSE370. The project brought together three core layers — a structured SQL database, a PHP backend, and an HTML/CSS frontend into a functional and runnable university management system.

References

CSE370 course textbook (Database System Concepts) and resources.

